

Business Recommendations

Customer Segmentation & Prediction Project

Three customer segments were identified via K-Means clustering on tenure, charges, contract type, payment method, paperless billing, and senior-citizen status. A dedicated Random Forest churn model was tuned and evaluated for each segment. This document translates those results into segment-specific retention strategies and a rough estimate of the revenue impact.

Segment Overview

Segment	Customers	Avg Tenure	Avg Monthly Charges	Churn Rate	Model F1 / ROC-AUC
Premium Loyalists	232	37.5 mo	\$122.87	10.0%	0.80 / 1.00
Budget Regulars	124	37.0 mo	\$107.28	15.0%	0.89 / 0.99
Standard Newcomers	144	34.6 mo	\$103.56	7.0%	0.67 / 0.94

Full profiles (contract mix, payment method, paperless billing) are in *segment_profiles.md*; full metrics in *model_evaluation_results.csv*.

Segment-Specific Strategies

Premium Loyalists — Convert flexibility into commitment

This is the largest and highest-spending segment, but 45% remain on flexible month-to-month terms rather than a long-term contract. Their churn rate (10.0%) tracks the overall base rate, so the opportunity here is proactive, not reactive.

- Offer a One-year or Two-year contract upgrade incentive (loyalty pricing or a service credit) targeted at the month-to-month customers in this segment.
- Use the trained model's churn-risk score to prioritize outreach to the highest-risk accounts within the segment rather than contacting everyone.
- Protect margin: these are the highest-ARPU customers, so retention spend per customer can be higher here than in other segments.

Budget Regulars — Highest churn risk, needs the most attention

Smallest segment but highest churn rate (15.0%), well above the 10.6% base rate. Uniformly pays by Credit Card and has lower average spend than the other two segments — this group should be the top priority for a retention program.

- Launch a targeted retention campaign for this segment first — it has the best ratio of at-risk customers to segment size.
- Investigate the Credit-Card-only payment pattern: confirm it's a genuine behavioral signal (e.g. billing friction) rather than a data artifact before designing an intervention around it.

- Test value-oriented offers (bundle discounts, usage-based add-ons) rather than premium upsells, given this segment's lower spend profile.

Standard Newcomers — Lowest churn, protect what's working

Shortest average tenure but the lowest churn rate (7.0%) of any segment, entirely attributable to universal Two-year contract adoption. This segment validates the contract-length strategy recommended for Premium Loyalists.

- Maintain the Two-year contract as the default offer for new customers acquired through whatever channel feeds this segment.
- Since churn is already low, prioritize retention spend elsewhere (Budget Regulars, Premium Loyalists) rather than over-investing here.
- Monitor this segment as it ages past its current short tenure — its churn rate under longer real-world tenure has not yet been observed.

Estimated Business Impact

Rough estimate only, assuming a targeted retention program reduces churn by 20% within each segment (a conservative, illustrative figure — not a guarantee). Calculated as: at-risk customers × average monthly charge × 20%.

Segment	At-Risk Customers	Monthly Revenue At Risk	Est. Monthly Revenue Protected
Premium Loyalists	23.2	\$2,708.37	\$541.67
Budget Regulars	18.6	\$1,995.41	\$399.08
Standard Newcomers	10.1	\$1,150.23	\$230.05
Total	51.9	\$5,854.01	\$1,170.80

At an illustrative 20% churn reduction, the combined program is estimated to protect roughly **\$1,171/month (~\$14,050/year)** in revenue across the three segments, with Premium Loyalists and Budget Regulars contributing the largest share.

Recommended Priority Order

- **1. Budget Regulars** — highest churn rate; smallest, most tractable segment for a pilot program.
- **2. Premium Loyalists** — largest revenue base and highest absolute revenue at risk; contract-upgrade incentive.
- **3. Standard Newcomers** — already performing well; monitor and maintain rather than actively intervene.

Note: figures are derived from a 500-customer sample and per-segment churn models trained on relatively few positive (churned) cases; treat the business impact estimate as directional rather than precise, and validate with a small pilot before a full rollout.